

Damping Elements

When a system vibrates, the vibrational energy is slowly dissipated in the form of sound or heat. The mechanism by which the vibrational energy is dissipated is known as damping. A damper is assumed to have neither mass nor elasticity. **The damping force exists only if there is relative velocity.**

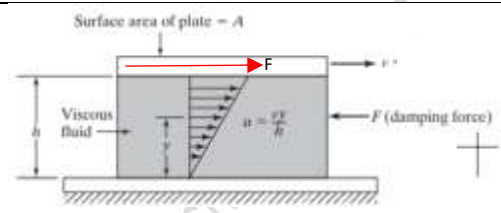
The damping is modelled as one or more of the following types.

Viscous damping

It is the most commonly used damping mechanism in vibration analysis. When a system vibrates in a fluid medium like air, gas, water, oil etc., the energy dissipation due to the fluid friction is called viscous damping. In such case, the dissipation of energy depends on factors like viscosity of fluid, size and shape of the vibrating body, frequency of vibration, velocity etc.

When a plate moves relatively with respect to another parallel plate with a viscous fluid in between, a viscous damper can be obtained.

Let a force F acting on the top plate moves it with a velocity v relative to the bottom plate.



From Newton's law of viscous flow,

$$\text{shear stress exerted on fluid, } \tau = \mu \frac{du}{dy} \ll \gg F = \mu A \frac{v}{h}$$

The resisting force acting on the plate, $F = \left(\frac{\mu A}{h}\right) v$. This is known as the damping force.

We can see that the damping force is proportional to the relative velocity. Usually, the damping force is written as $F_d = c\dot{x}$, where c is a constant known as damping coefficient.

The energy dissipated per cycle due to viscous damping,

$$W_d = \pi c \omega X^2 \ll \gg \omega \rightarrow \text{frequency, } X \rightarrow \text{amplitude}$$

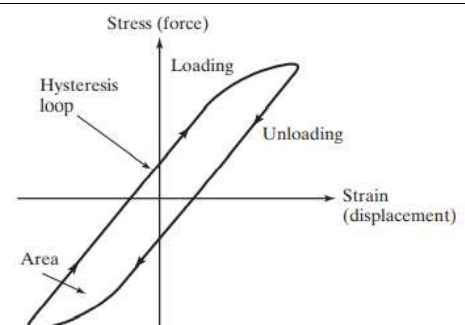
Coulomb damping

In this case, the damping is caused by the dry friction existing between the surfaces having relative motion. Here, the damping force is constant in magnitude but acts in a direction opposite to the direction of motion of the body.

Material/Hysteretic damping

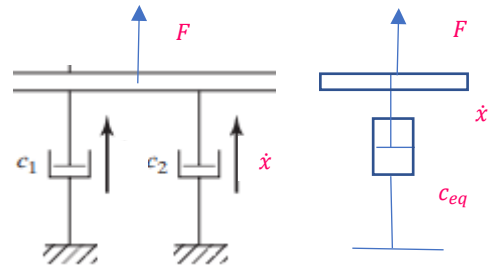
When a material is deformed, energy is dissipated by the material due to the friction between the atomic planes

When a body having material damping is subjected to vibration, the stress strain diagram exhibits the hysteresis loop as shown. The area of the hysteresis loop is equal to the energy dissipated per unit volume of the body per cycle due to damping.



Example 1

Consider two viscous dampers in parallel. Find the equivalent damping coefficient.



Since both the dampers are parallel, the displacement and hence the velocities are equal. But, the forces are different.

$$F = F_1 + F_2 = c_1 \dot{x} + c_2 \dot{x}$$

For the equivalent damper, $F = c_{eq} \dot{x}$

$$\therefore c_{eq} \dot{x} = c_1 \dot{x} + c_2 \dot{x} \ggg c_{eq} = (c_1 + c_2)$$

The energy dissipated by the two dampers,

$$W_d = \pi c_1 \omega X^2 + \pi c_2 \omega X^2 = \pi (c_1 + c_2) \omega X^2$$

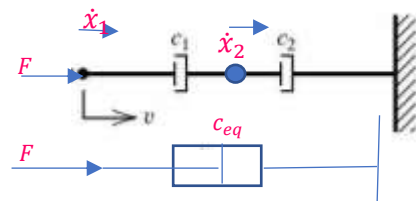
If these dampers are replaced by an equivalent damper of damping coefficient c_{eq} . The energy dissipated by the equivalent damper is given by $W_{eq} = \pi c_{eq} \omega X^2$

For both systems to be equivalent,

$$W_{eq} = W_d \ggg \pi c_{eq} \omega X^2 = \pi (c_1 + c_2) \omega X^2 \lll c_{eq} = (c_1 + c_2)$$

Example 2

Consider two viscous dampers in series. Find the equivalent damping coefficient.



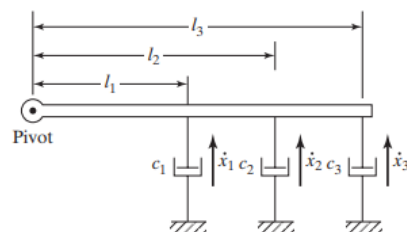
Here, the same force is acting on both the dampers. Net velocity, $v = \dot{x}_1 + \dot{x}_2 = \frac{F}{c_1} + \frac{F}{c_2}$

For the equivalent damper, $v = \frac{F}{c_{eq}}$

$$\therefore \frac{F}{c_{eq}} = \frac{F}{c_1} + \frac{F}{c_2} \ggg \frac{1}{c_{eq}} = \frac{1}{c_1} + \frac{1}{c_2}$$

Exercise 1

Determine the equivalent damping coefficient of the system, assuming it to be connected at the location of c_3

**Reference:**

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition